JEE Mains 2020 Jan Chapter wise Question Bank

Mathematical Reasoning

Q1

If f(x) is continuous and differentiable in $x \in [-7, 0]$ and $f'(x) \le 2\ \forall x \in [-7, 0]$, also $f(-7) = -3$ then range of $f(-1) + f(0)$

यदि f(x), $x \in [-7, 0]$ में सतत् तथा अवकलनीय है तथा $f'(x) \le 2\ \forall x \in [-7, 0]$ और $f(-7) = -3$ तो $f(-1) + f(0)$ का परिसर है–

(1) $[-5, -7]$ (2) $(-\infty, 6]$ (3) $(-\infty, 20]$ (4) $[-5, 3]$

7th Jan Morning

Sol

(3)

Lets use LMVT for $x \in [-7, -1]$ में लांग्राज मध्यमान प्रमेय से

$$\frac{f(-1) - f(-7)}{(-1+7)} \le 2$$

$$\frac{f(-1) + 3}{6} \le 2 \Rightarrow f(-1) \le 9$$

Also use LMVT for $x \in [-7, 0]$ में लांग्राज मध्यमान प्रमेय से

$$\frac{f(0) - f(-7)}{(0+7)} \le 2$$

$$\frac{f(0)+3}{7} \le 2 \Rightarrow f(0) \le 11 \qquad \therefore \qquad f(0) + f(-1) \le 20$$

02

Contrapositive of if $A \subset B$ and $B \subset C$ then $C \subset D$

(1) $C \not\subset D$ or $A \not\subset B$ or $B \not\subset C$

(2) $C \subset D$ and $A \not\subset B$ or $B \not\subset C$

(3) $C \subset D$ or $A \not\subset B$ and $B \not\subset C$

(4) $C \subset D$ or $A \not\subset B$ or $B \not\subset C$

7th Jan Evening

Sol

Let $P = A \subset B$, $Q = B \subset C$, $R = C \subset A$

Contrapositive of $(P \wedge Q) \rightarrow R$ is $\sim R \rightarrow \sim (P \wedge Q)$

$R \vee (\sim P \vee \sim Q)$

03

Which of the following is tautology

निम्न में से कौनसी पुनःरुक्ति है ?

(1) $(p \wedge (p \to q)) \to q$

(2) $q \to p \wedge (p \to q)$

(3) $p \vee (p \wedge q)$

(4) $(p \wedge (p \vee q)$

8th Jan Morning

Sol

(1)

p	q	p→ q	p∧ (p → q)	(p ∧(p →q)) → q	q → p∧(p→q)	p ∧ q	p ∨ (p ∧q)	p ∨ q	p∧ (p∨ q)
T	T	T	T	T	T	T	T	T	T
T	F	F	F	T	T	F	T	T	T
F	T	T	F	T	F	F	F	T	F
F	F	T	F	T	T	F	F	F	F

04

Which of the following is tautology

निम्न में से कौनसी पुनःरुक्ति है ?

(1) $(p \wedge (p \to q)) \to q$

(2) $q \to p \wedge (p \to q)$

(3) $p \vee (p \wedge q)$

(4) $(p \wedge (p \vee q)$

8th Jan Evening

Sol

(1)

$(\sim p \wedge q) \to (p \vee q)$

$\sim\{(\sim p \wedge q) \wedge (\sim p \wedge \sim q)\}$

$\sim\{\sim p \wedge f\}$

05

The negation of '$\sqrt{5}$ is an integer or 5 is an irrational number' is

(1) $\sqrt{5}$ is an integer and 5 is not an irrational Number

(2) $\sqrt{5}$ is not an integer and 5 is an irrational Number

(3) $\sqrt{5}$ is not an integer or 5 is not an irrational Number

(4) $\sqrt{5}$ is not an integer and 5 is not an irrational Number

9th Jan Morning

Sol

(4)

$\sqrt{5}$ is not an integer and 5 is not an irrational Number $\sim(p \vee q) = \sim p \wedge \sim q$

06

If $p \rightarrow (p \wedge \sim q)$ is false. Truth value of p & q will be

यदि $p \rightarrow (p \wedge \sim q)$ असत्य है, तब p & q का सत्यतता मान होगा

(1) TT (2) TF (3) F T (4) F F

9th Jan Evening

Sol

(1)

p	q	~q	p∧~q	p→(p∧~q)
T	T	F	F	F
T	F	T	T	T
F	T	F	F	T
F	F	T	F	T